Grade 7

DATA HANDLING

WHAT IS DATA?

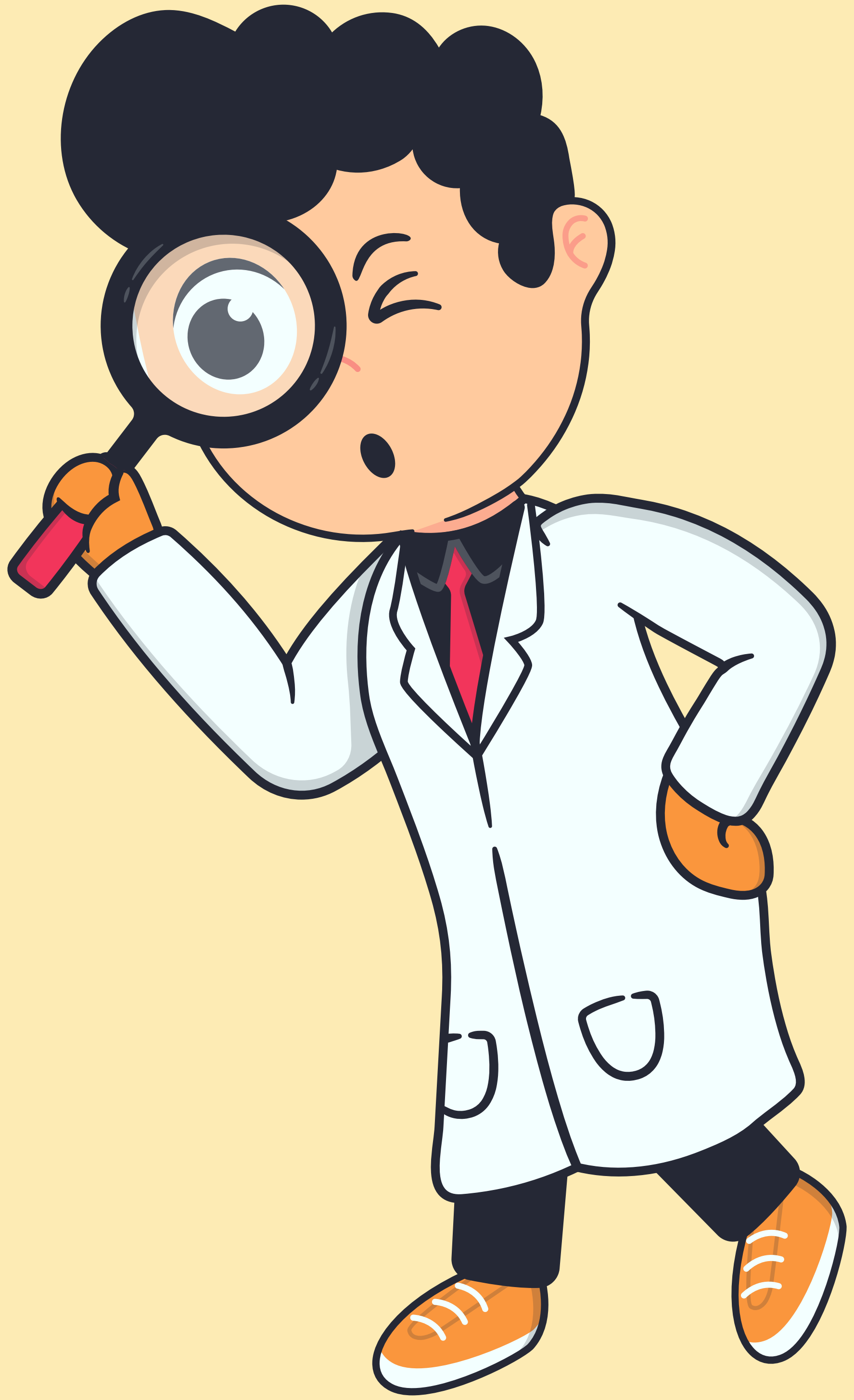
Data describes any facts or statistics which can be collected. Data is collected a lot in science! Data can be either:

- Quantitative - Data that is described in numbers.
- Qualitative - Data that is described in words.

WHAT IS SCIENCE DATA?

Science data is the information collected during experiments and investigations.

It is used to answer question and understand the world around us.



COLLECTING RAW DATA

Raw data is the original, unedited information collected during an experiment.

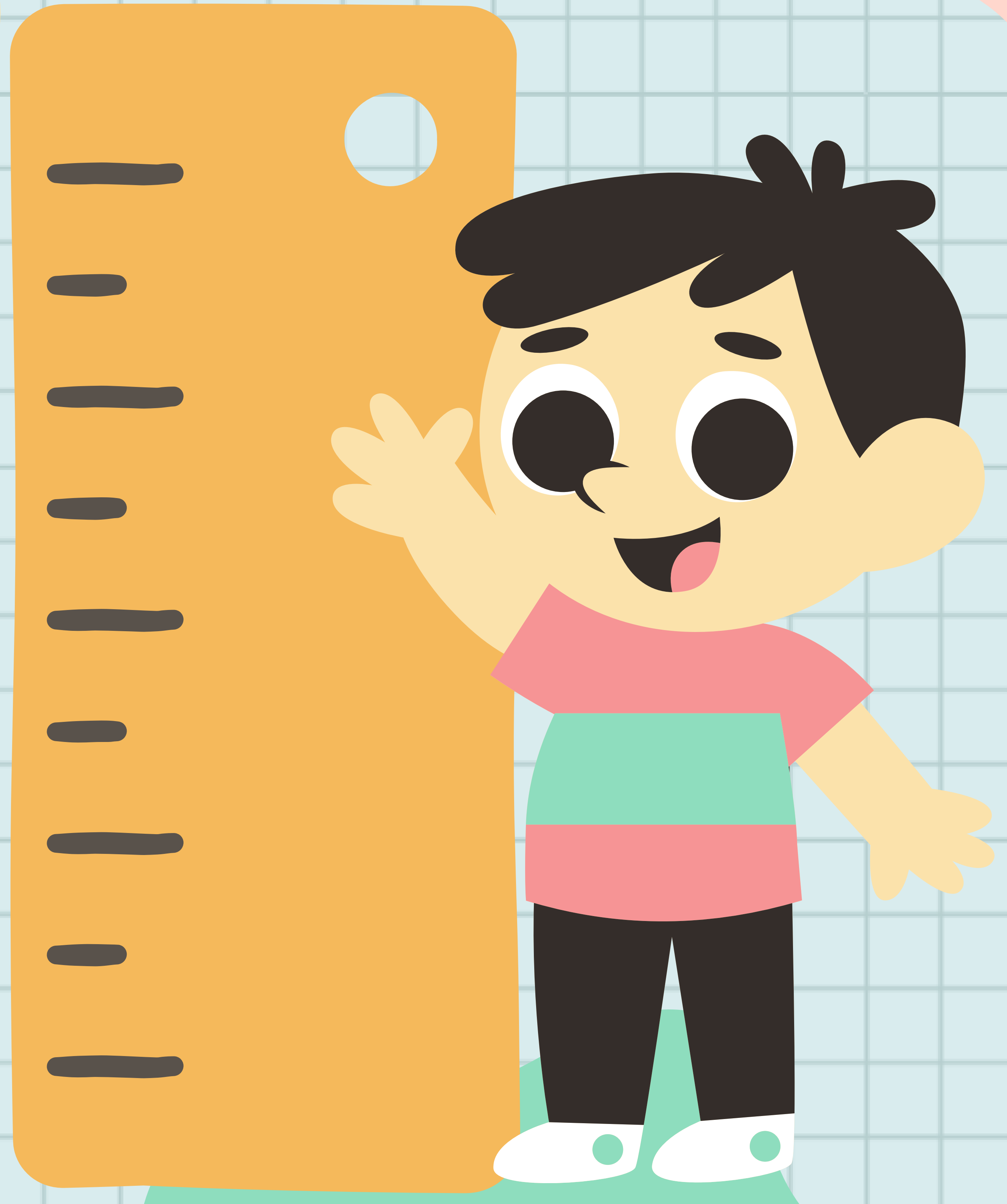
It includes only information that is observed or measured - no changes or calculations

Example: Plant A grew 3.1 cm
Plant B grew 2.8 cm
Plant C grew 3.5 cm

CONTINUOUS DATA

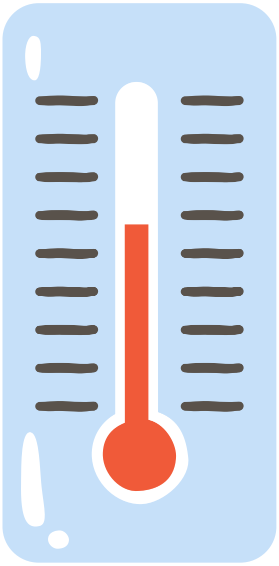
Continuous data is numeric data that can have any value within a certain range.

Examples of continuous data include, height, temperature and time.



CONTINUOUS DATA EXAMPLES

Time



Temperature

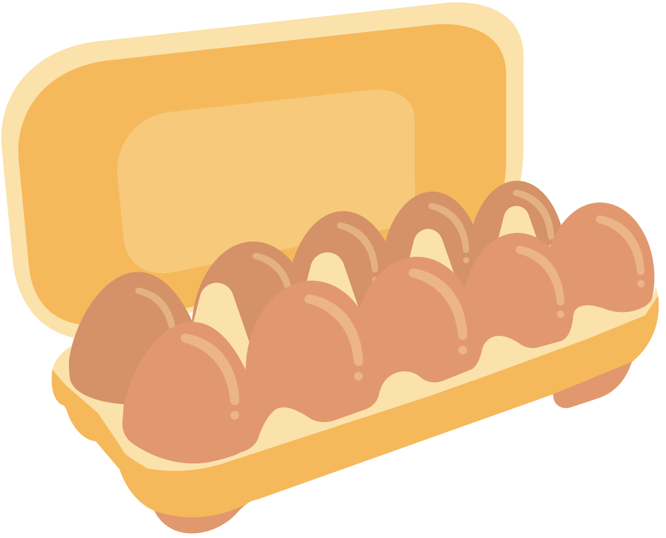
Weight

DISCRETE DATA

Discrete data is numeric data that can only have certain values.

Examples of discrete data include, shoe size, numeric clothes sizes and the number of people sat down on a bus.

DISCRETE DATA EXAMPLES



Number of eggs in a carton

Number of pages in a book

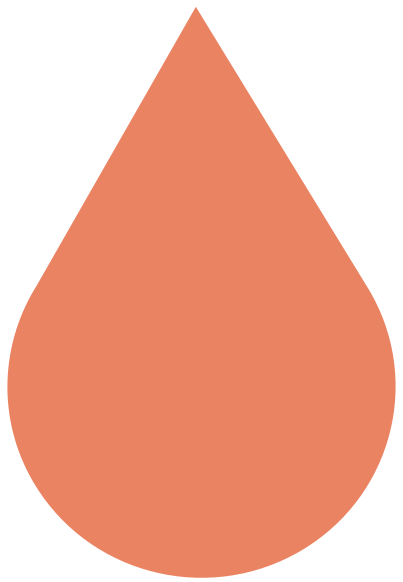
Numerical clothes sizes

CATEGORIC DATA

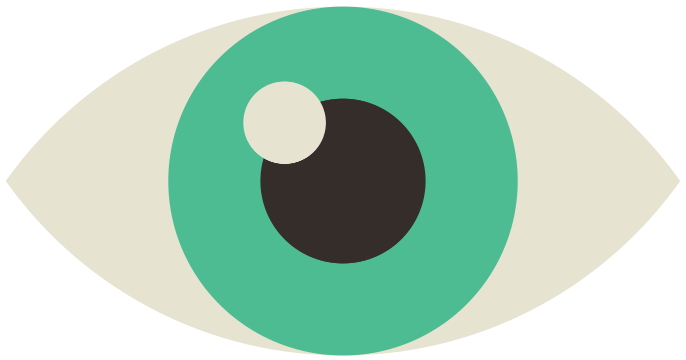
Categoric data is in word form and describes data which fits into a specific group or category.

Examples of categoric data include, eye colour, shape, blood type and nationality.

CATEGORIC DATA EXAMPLES



Blood type



Eye colour

Hair colour

CHECK FOR UNDERSTANDING #1

Identify each of the following pieces of data about the girl pictured to the right as either qualitative or quantitative:

1. Red pants
2. Short hair
3. Five feet, four inches tall
4. Seventeen years old
5. Outgoing personality

CHECK FOR UNDERSTANDING #1

Identify each of the following pieces of data about the girl pictured to the right as either qualitative or quantitative:

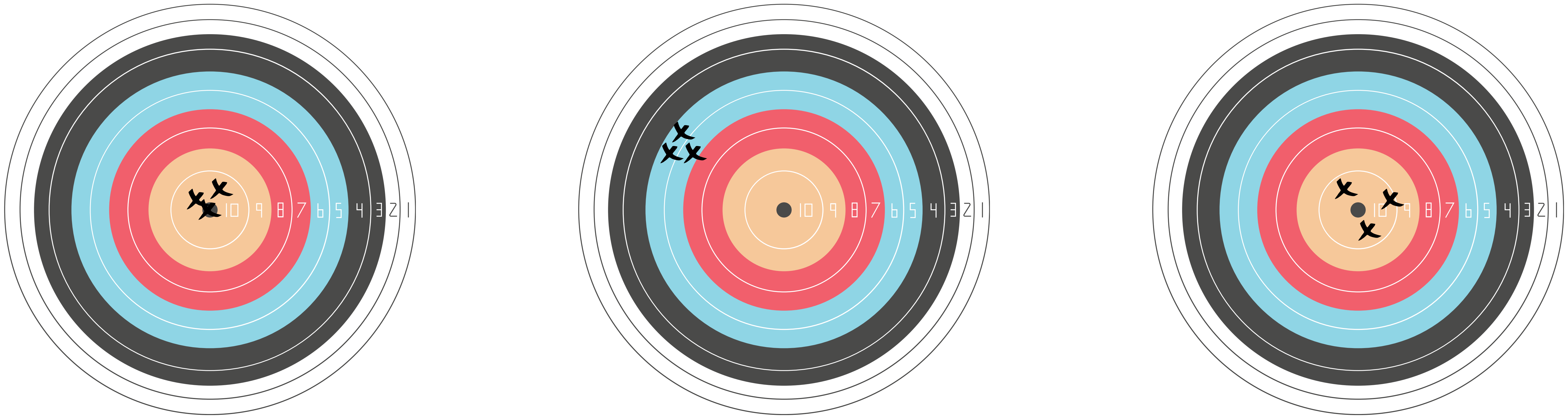
1. Red pants = **qualitative**
2. Short hair = **qualitative**
3. Five feet, four inches tall = **quantitative**
4. Seventeen years old = **quantitative**
5. Outgoing personality = **qualitative**

EXPERIMENTAL DATA

When designing an experiment, you must decide what type of data you will collect. This is related to your **dependent variable**. It should be the goal of all researchers to report data that is both **accurate** (matching known results) and **reliable** (matching other experimental results), no matter what type of data it is.

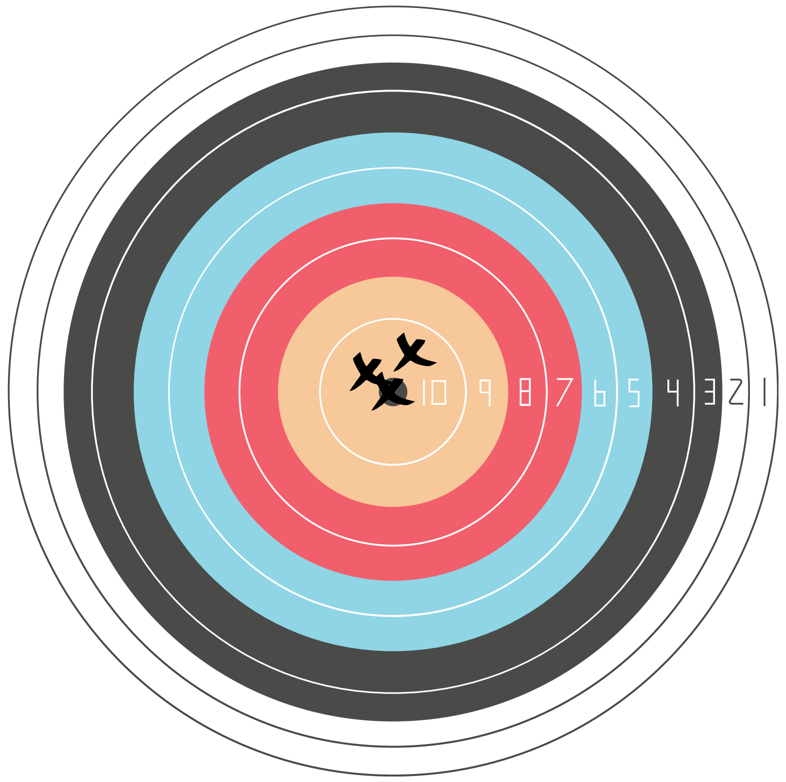
CHECK FOR UNDERSTANDING #2

Which of the targets below displays both accurate and reliable aims? Which displays only accurate aim? Only reliable aim?

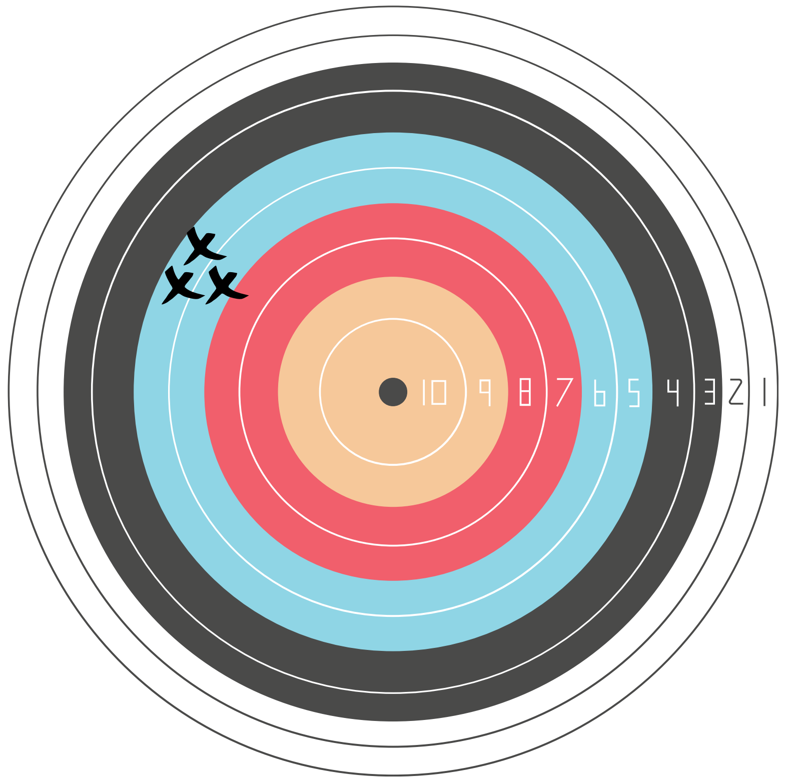


CHECK FOR UNDERSTANDING #2

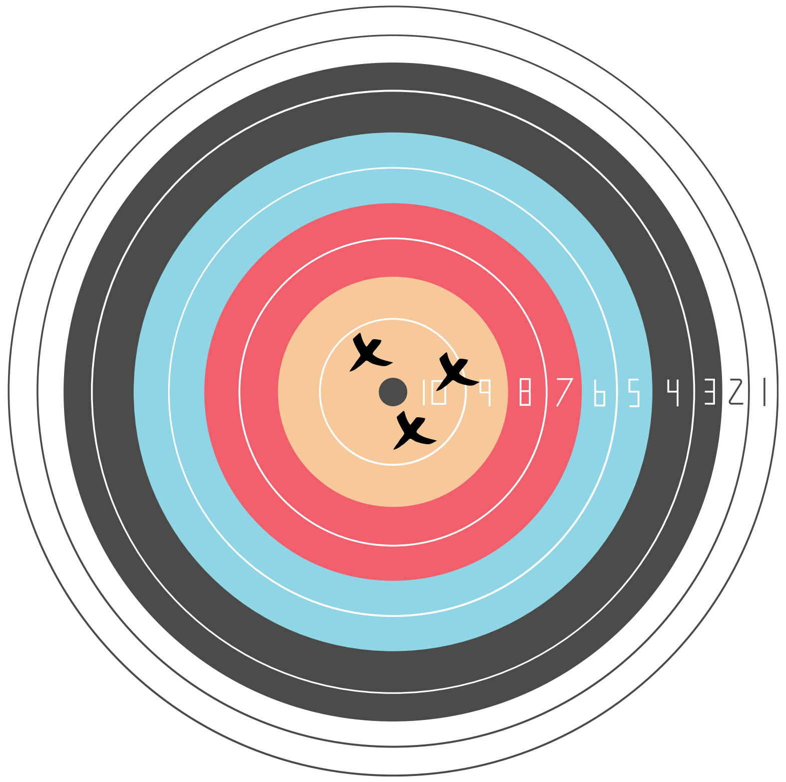
Accurate
& Reliable



Reliable

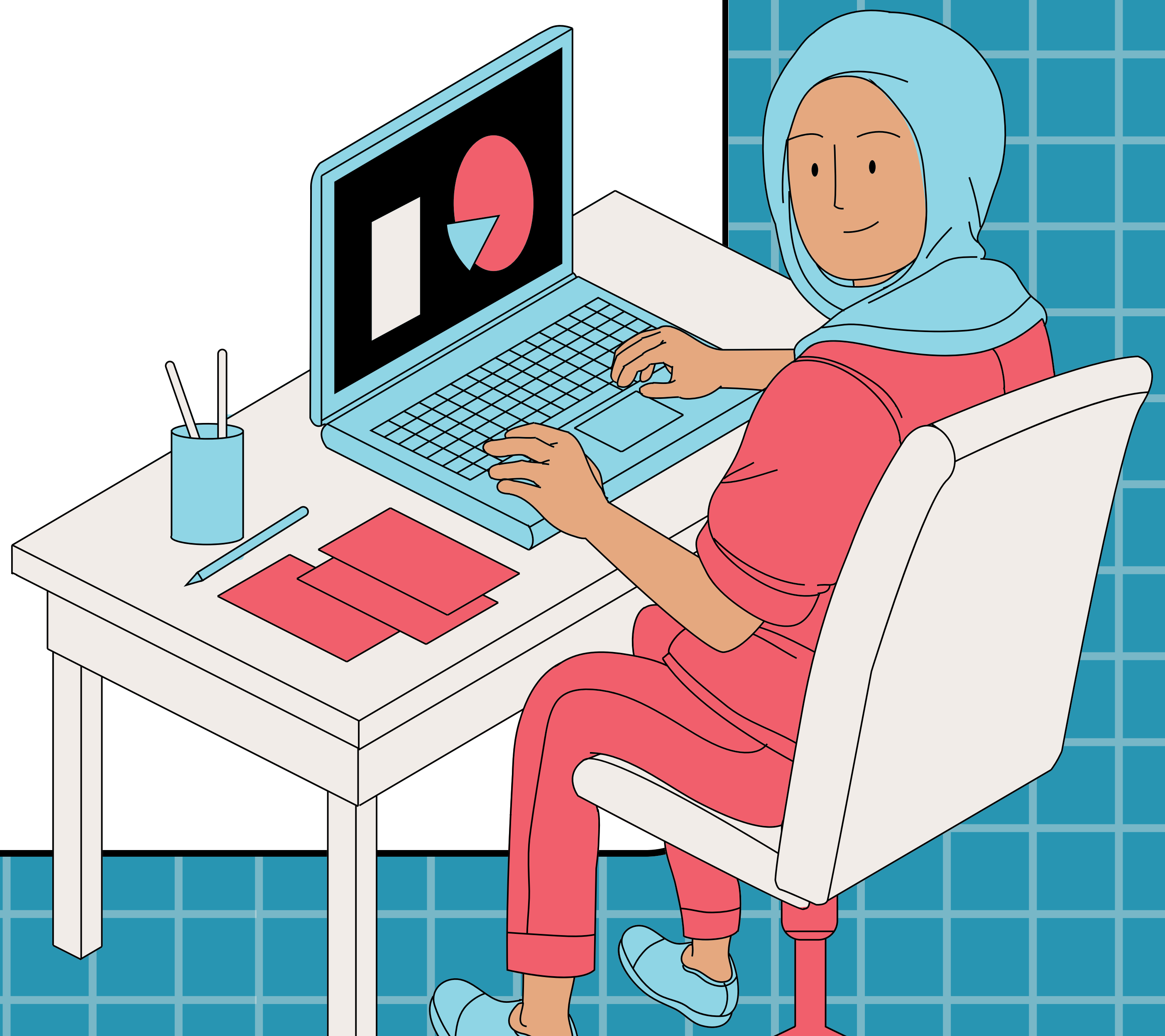


Accurate



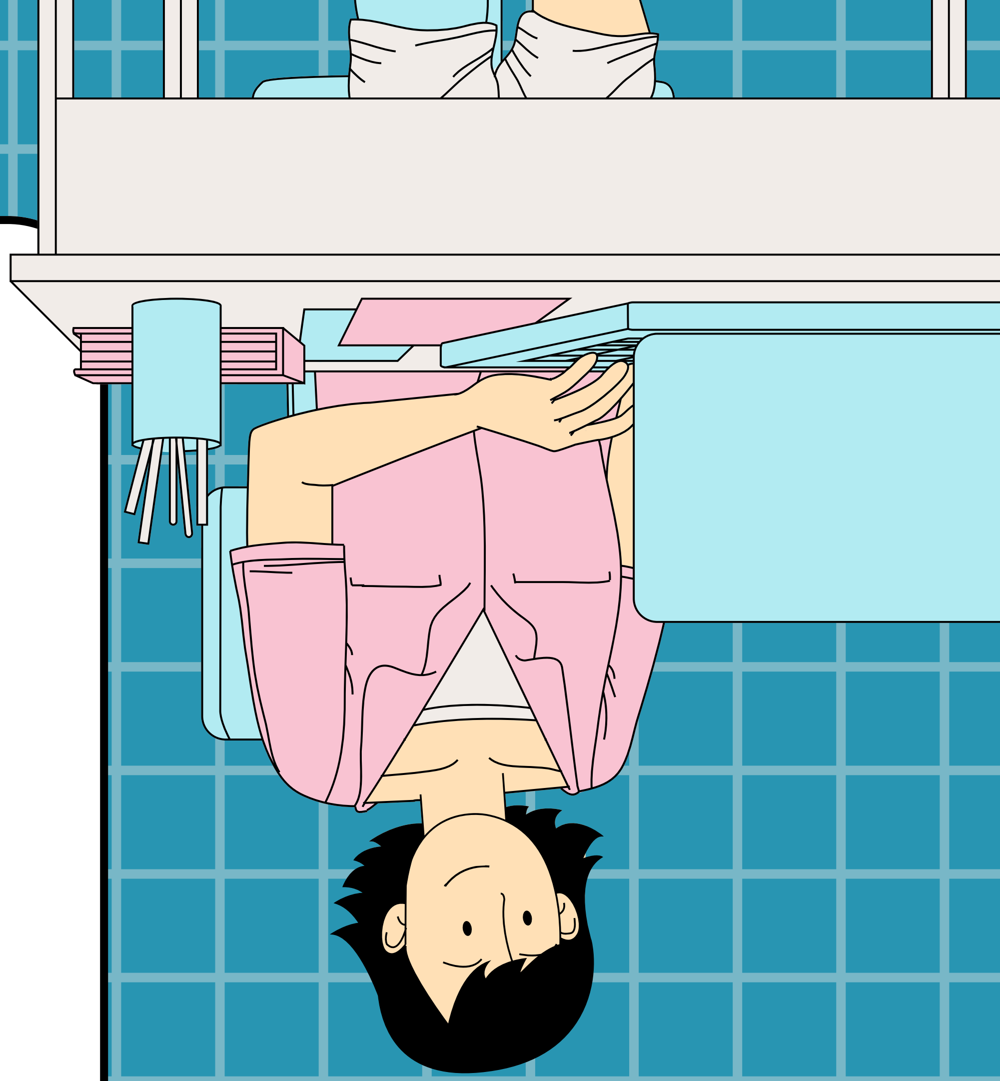
ACCURACY & RELIABILITY

You can increase the accuracy and reliability of your data and decrease **error** by ensuring you have an appropriate **sample size** (the number of subjects being tested during an experiment) and making your experiment **repeatable** so you can gain more data points when you conduct it again.



DATA TABLES

During an experiment, data is typically collected and organized using data tables with the independent variable on the left side and the dependent variable on the right side -- with units included!



BASIC STATISTICS

While data tables are great for organizing data, they are not always easy to interpret, especially when it comes to large data sets. This is why we use statistics to help us understand data. There are three basic statistics: mean, median, and mode. These values help to reveal what is 'typical' in a data set.

HOW TO PROCESS DATA

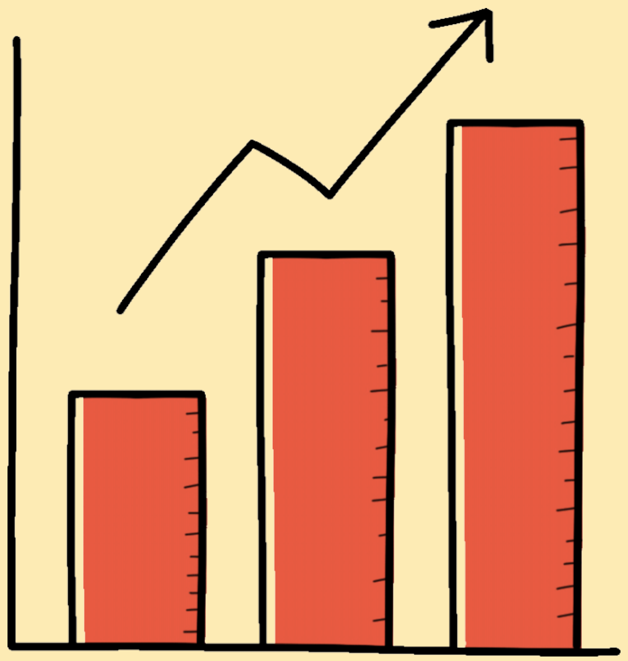
1. Check for mistakes in the raw data
2. Organize the data in a table or chart
3. Calculate averages, differences, totals, ect
4. Represent the numbers visually in a graph
5. Look for patterns and trends
6. Explain the data using scientific reasoning

PROCESSED DATA EXAMPLE

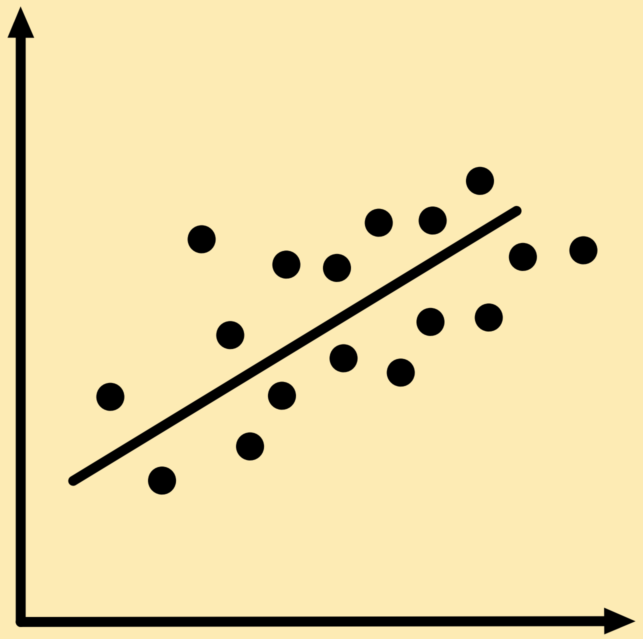
Plant	Plant Height (cm)				Total Plant Growth (cm)
	Day 1	Day 5	Day 10	Day 15	
A	5.0	6.1	6.9	8.1	3.1
B	5.0	6.5	7.4	9.2	4.2
C	5.0	6.3	7.3	9.6	4.6

The raw data, measuring the plant's height each day, was processed, by using the information to find the total growth of the plant over time.

GRAPHING PROCESSED DATA



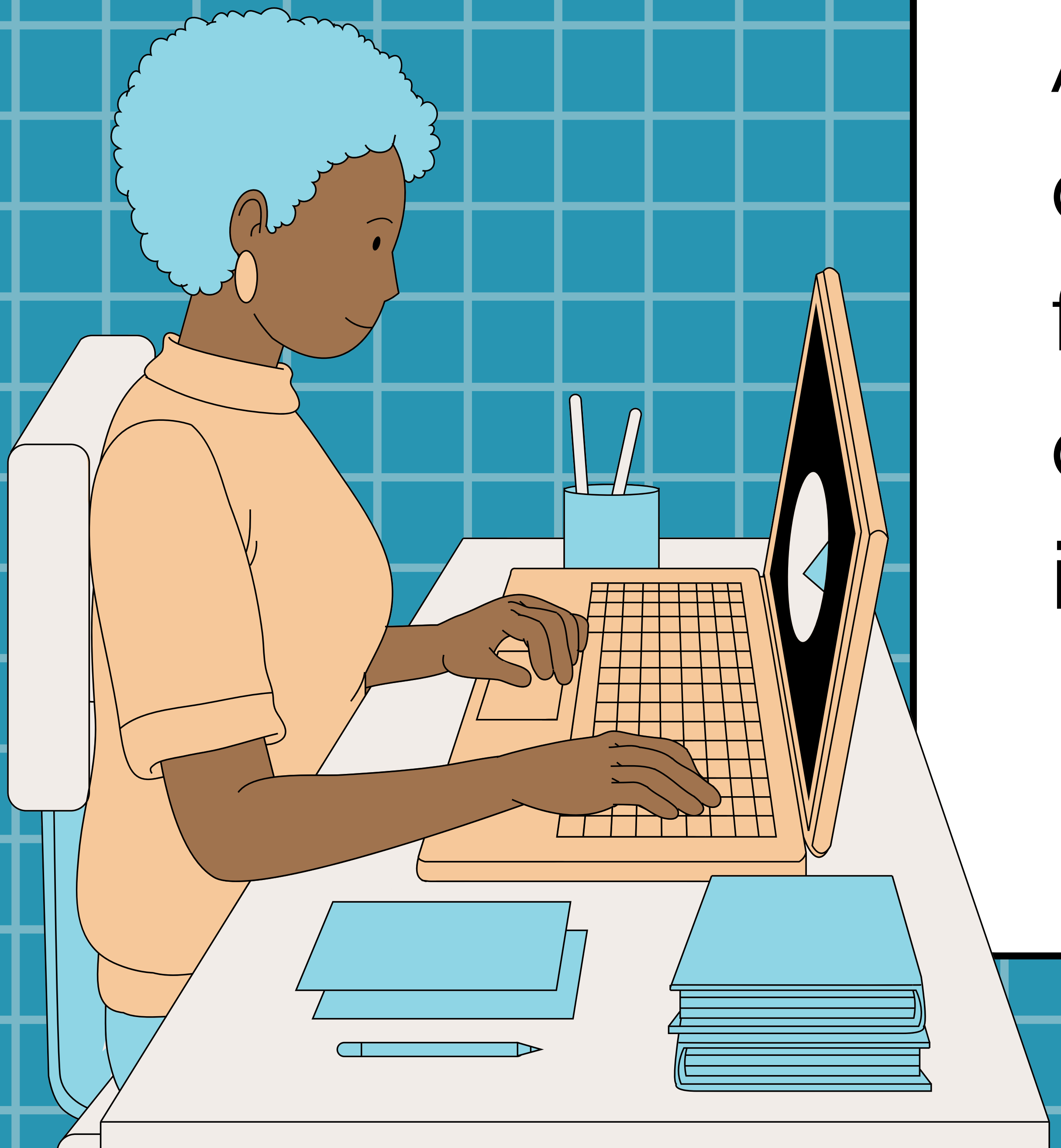
A bar graph is used to compare different groups or categories.



A pie graph is used to show how something changes over time. It is used to help you see trends and patterns.

A pie graph is used to show parts of a whole. It helps to see how something is divided into pieces or percentages.

CALCULATING MEAN



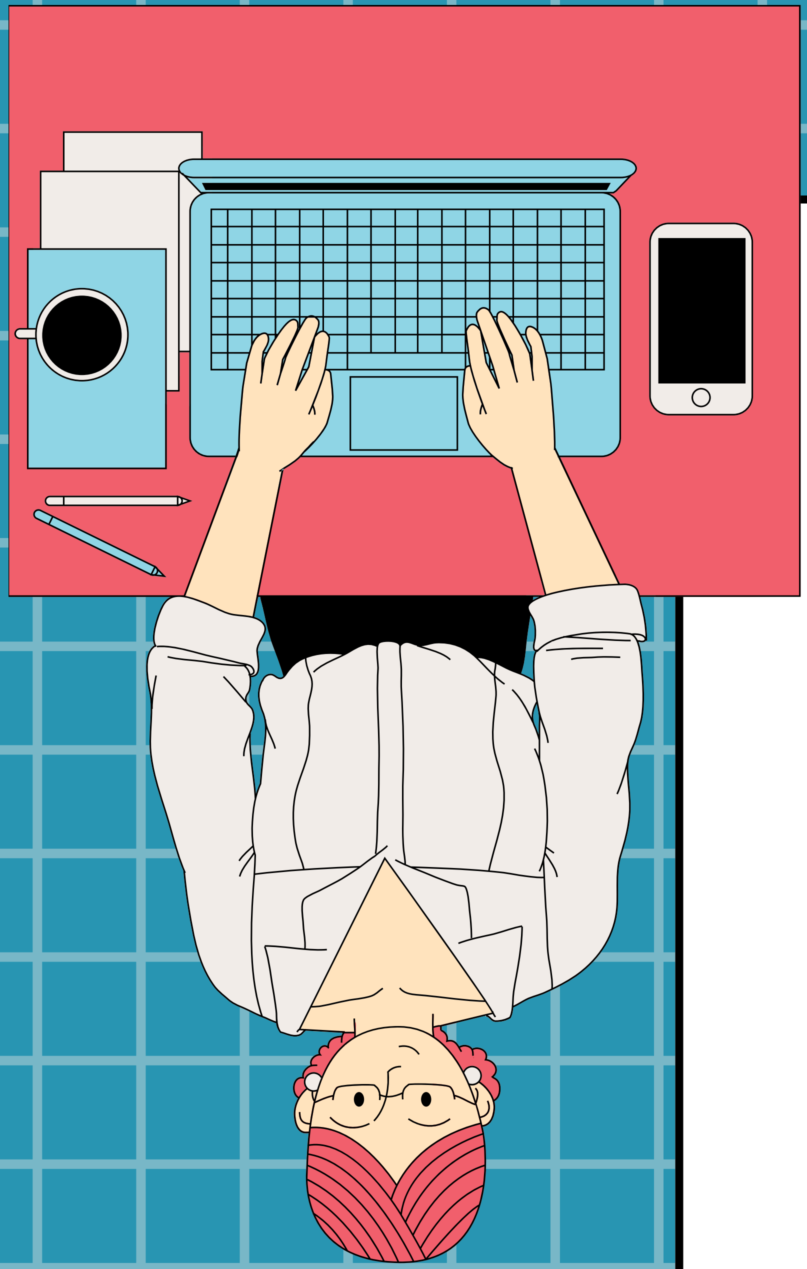
A **mean** is another word for an average. To determine the average of a set of values, first, find the sum of all the numbers. Then, divide the sum by the total number of values in the set.

Example: (a + b + c) /3

CALCULATING MEDIAN

A **median** is the middle value in a set of numbers. To find the median, arrange all of the values in order from lowest to highest. Then, find the middle value. If there is no single middle number, you can find the average of the two middle numbers.

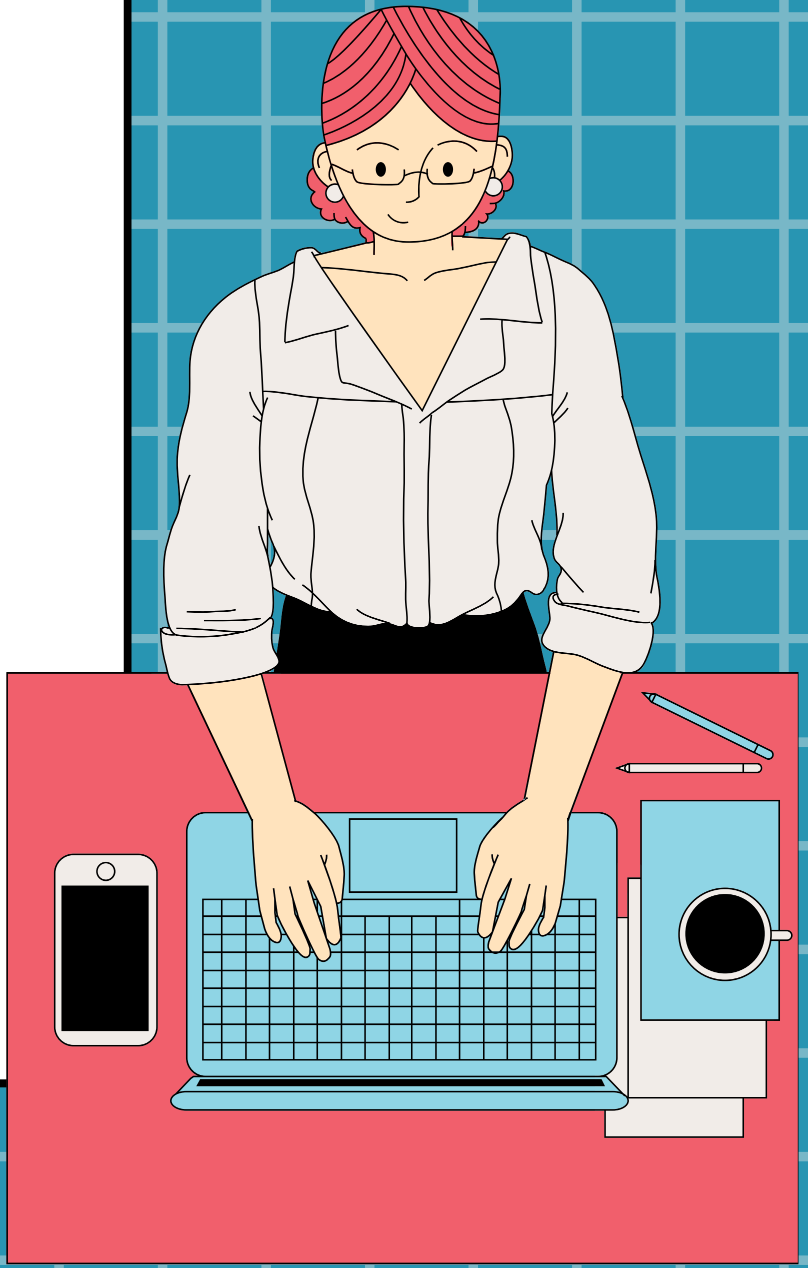
Example: a b <u>c</u> d e



CALCULATING MODE

A **mode** is a value that occurs most often in a data set. To find the mode, simply look for the number that is repeated the most. Some data sets do not have a mode, whereas others have more than one mode.

Example: a a b c d



CHECK FOR UNDERSTANDING #3

A researcher is conducting an experiment on the effects of music on plant growth. She has multiple groups of plants composed of five plants each. On the 7th day of her experiment, she measures the following heights for Group 1 of her plants: 9 in., 11 in., 9 in., 13 in., and 8 in. Find the mean, median, and mode of this data set.

CHECK FOR UNDERSTANDING #3

A researcher is conducting an experiment on the effects of music on plant growth. She has multiple groups of plants composed of five plants each. On the 7th day of her experiment, she measures the following heights for Group 1 of her plants: 9 in., 11 in., 9 in., 13 in., and 8 in. Find the mean, median, and mode of this data set.

Mean: 10 in. **Median: 9 in.** **Mode: 9 in.**

The End of the Presentation

Thanks for listening!